Coronary circulation

- Heart is supplied right and left coronary arteries
- Right coronary artery is dominant in 50%
- Left coronary artery is dominant in 20%

Special features of Coronary Circulation

1. Blood flow 250ml/min

2. Heart receives major blood supply during diastole

- Blood vessels compressed during systole due to contraction of ventricular muscle

- Decrease in blood supply to muscle

3.Coronary arteries are end arteries

- Blockage of coronary artery results in ischemia and infarction of cardiac muscle

4. Adequate coronary blood flow reserve : coronary blood flow can be increased about 4-5 folds to meet heart more oxygen during exercise

5.Main source of energy for heart muscle is free fatty acids

6. Heart muscle extracts 80% oxygen from arterial blood even at rest : Arteriovenous difference is higher

7. Autoregulation : coronary blood flow is constant within pressure range of 70-110mmHg

Arterial Pulse

Arterial Pulse

- It is defined as rhythmic expansion of the arterial wall due to transmission of pressure waves along the walls of the arteries during each systole of heart

- Clinically radial pulse is examined for assessment of arterial pulse

- Normally heart rate is 60 to 100 beats per minute